RESEARCH HIGHLIGHTS

Autonomy at the Edge: Governance, Risk, and International Policy Frameworks for AI-Driven Deep Space Communications

Jason Pandian[1] and I. Kala[2]

[1]Department of Information Technology, Nehru Institute of Technology, India

[2]Department of Computer Science and Engineering, PSG Institute of Technology and Applied Research, India

---

- **Spectrum Policy Gaps:** Analyzes acute policy gaps in ITU regulations regarding dynamic, space-grade AI spectrum allocation and the Outer Space Treaty's access principles.

- **Physical Infrastructure Asymmetries:** Exposes geopolitical asymmetries in deep space ground network infrastructure and proprietary software control (NASA, ESA, CNSA, ISRO, and JAXA complexes).

- **Algorithmic Bias Modeling:** Models a multinational Martian dust storm scenario to demonstrate systemic bias in autonomous "data triage" loops.

- **Explainable AI (XAI) Framework:** Proposes an auditable, cryptographic Explainable AI (XAI) framework for spacecraft downlink priority and write-once decision logs.

- **COPUOS Code of Practice:** Drafts a concrete "International Code of Practice for Autonomy in Deep Space Telecommunications" under UN COPUOS to establish algorithmic transparency and dynamic envelopes.

GRAPHICAL ABSTRACT

Autonomy at the Edge: Governance, Risk, and International Policy Frameworks for AI-Driven Deep Space Communications